Supplementary Material S2. Records Excluded at Full-Text Assessment (n = 131)

Records are numbered as in the database search results (see S1). Each record was assigned the primary reason for exclusion.

Database	No.	Author(s)	Prediction task	Exclusion reason
Scopus	3	Han et al. (2026)	MPC operating-cost/tail-risk surrogate prediction and risk-adaptive scheduling	Prediction target not carbon-related
Scopus	6	Khorshidi et al. (2026)	28-day compressive-strength prediction and multi-objective mix optimization	Prediction target not carbon-related
Scopus	7	Yan (2026)	Energy Consumption Forecasting (Regression)	Prediction target not carbon-related
Scopus	9	Tang (2026)	Energy Price Index and Renewable Energy Share forecasting (Regression)	Prediction target not carbon-related
Scopus	10	Keshireddy et al. (2026)	Combustion Performance Forecasting (Efficiency, CO Emissions, Fuel Consumption Index)	Prediction target not carbon-related
Scopus	11	Elmousalami and Alotaibi (2026)	Project budget, cost variance, duration, duration variance and project-risk prediction	Prediction target not carbon-related
Scopus	12	Dewangan et al. (2026)	Concrete property prediction (slump, flow, compressive strength)	Prediction target not carbon-related
Scopus	13	Masmali et al. (2026)	Sustainability scoring, anomaly/ESG violation detection, audit decisions; carbon-intensity forecasting as a separate green-intelligence component	Carbon emissions derived indirectly or not the XAI-explained target
Scopus	14	Ji et al. (2026)	Heating and cooling energy use intensity (EUI) prediction across climate zones	Prediction target not carbon-related
Scopus	15	Datta et al. (2026)	Compressive-strength prediction	Prediction target not carbon-related
Scopus	18	Duan et al. (2026)	NH_3 and H_2S emissions prediction	Prediction target not carbon-related
Scopus	19	Li et al. (2026)	Provincial total energy consumption prediction and low-carbon policy analysis	Prediction target not carbon-related
Scopus	20	Khare and Bhatia (2026)	Sustainable building governance and net-zero readiness forecasting	Prediction target not carbon-related
Scopus	21	Wang et al. (2026)	Unconfined compressive strength (UCS) prediction	Prediction target not carbon-related
Scopus	22	Zhang et al. (2026)	Leaf net photosynthetic-rate prediction	Prediction target not carbon-related
Scopus	26	Gallego Fernández et al. (2026)	Fuel-flow prediction and indirect reconstruction of combustion CO_2 emissions	Carbon emissions derived indirectly or not the XAI-explained target
Scopus	30	Paul et al. (2026)	Methane yield and methane emissions prediction	Carbon emissions derived indirectly or not the XAI-explained target
Scopus	31	Ban (2026)	Sustainability score prediction and resource allocation optimization	Prediction target not carbon-related
Scopus	34	Yan et al. (2026)	Coal blending ratio prediction	Prediction target not carbon-related
Scopus	35	Nageshwaran et al. (2026)	28-day compressive-strength prediction	Prediction target not carbon-related
Scopus	36	Das et al. (2026)	Outlet temperature prediction of PTC	Prediction target not carbon-related
Scopus	37	Verma et al. (2026)	Environmental compliance-status classification	Prediction target not carbon-related
Scopus	38	Gupta et al. (2026)	Environmental compliance classification and regulatory-risk prediction	Prediction target not carbon-related
Scopus	39	Wankhade et al. (2026)	Review and conceptual AI decision-support framework covering pollution, waste, water and carbon-emission monitoring	Review or non-original article
Scopus	43	Iftikhar et al. (2026)	Flexural strength prediction of cement-free plastic–sand composites	Prediction target not carbon-related
Scopus	46	Li et al. (2026)	Unit grouting amount prediction	Prediction target not carbon-related
Scopus	49	Hayawi et al. (2026)	Cooling load forecasting	Prediction target not carbon-related
Scopus	52	Atique et al. (2026)	Combined MPG prediction and CO_2 estimation from fuel consumption	Carbon emissions derived indirectly or not the XAI-explained target
Scopus	53	Wang et al. (2026)	Agricultural carbon emission efficiency (ACEE) forecasting	Carbon emissions derived indirectly or not the XAI-explained target
Scopus	55	Zhang et al. (2026)	Diagnostic analysis of street vitality–low-carbon coordination	Prediction target not carbon-related
Scopus	56	Franck-stève et al. (2026)	Electricity load forecasting	Prediction target not carbon-related
Scopus	57	Mirzaei et al. (2026)	28-day control-relative compressive-strength ratio prediction	Prediction target not carbon-related
Scopus	58	Reina-Jiménez et al. (2026)	Forecasting model selection and performance benchmarking	Prediction target not carbon-related
Scopus	60	Adamu et al. (2026)	Compressive strength prediction	Prediction target not carbon-related
Scopus	62	Rawat et al. (2026)	Phishing and impersonation detection	Prediction target not carbon-related

Database	No.	Author(s)	Prediction task	Exclusion reason
Scopus	63	Katlay and Turk (2026)	Compressive strength modeling	Prediction target not carbon-related
Scopus	65	Selicati et al. (2026)	Optimal EV charging schedule prediction	Prediction target not carbon-related
Scopus	66	Lee et al. (2026)	Hydrogen flow rate prediction	Prediction target not carbon-related
Scopus	67	Liu et al. (2026)	Operational electricity consumption forecasting	Prediction target not carbon-related
Scopus	68	Lotun et al. (2026)	Indoor heat gain prediction	Prediction target not carbon-related
Scopus	69	Wang et al. (2026)	Review of solar nowcasting methods	Review or non-original article
Scopus	77	Shi et al. (2026)	Yellowfin tuna fishing grounds prediction	Prediction target not carbon-related
Scopus	79	Marijan et al. (2026)	Review of fuel consumption estimation and optimization methods	Review or non-original article
Scopus	81	Mohsen and Mohsen (2026)	CH_4 and N_2O emissions prediction from manure management	No XAI applied
Scopus	82	Alatefi et al. (2026)	CO_2 uptake prediction in porous liquids	Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)
Scopus	84	Ezzat et al. (2026)	Average temperature rise prediction from agri-food CO_2 emissions	Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)
Scopus	85	Sungheetha et al. (2026)	Psilocybin treatment response and optimization modeling	Prediction target not carbon-related
Scopus	89	Zhang et al. (2026)	Concrete resource mismatch and daily concrete consumption prediction	Prediction target not carbon-related
Scopus	91	Aljubairi et al. (2026)	Energy-intensity prediction (kJ/pax-km) and policy scenario analysis	Prediction target not carbon-related
Scopus	92	Sheng et al. (2026)	Compressive-strength prediction under carbonation curing	Prediction target not carbon-related
Scopus	93	Liu et al. (2026)	28-day unconfined compressive strength (UCS) prediction	Prediction target not carbon-related
Scopus	96	Guo and Jin (2026)	Formic acid yield prediction	Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)
Scopus	101	Dong et al. (2026)	CO_2 adsorption, CO_2/N_2 selectivity, and CO_2/CH_4 selectivity prediction	Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)
Scopus	103	Su et al. (2026)	UCS prediction and mix proportion optimization	Prediction target not carbon-related
Scopus	104	Nguyen et al. (2026)	Port congestion and relative vessel fuel-demand prediction	Prediction target not carbon-related
Scopus	105	Ceken and Tuncal (2026)	Review of AI applications for fuel efficiency, emissions, maintenance, and aviation sustainability	Review or non-original article
Scopus	106	Nyberg et al. (2026)	PA voltage-level classification and PRB utilization forecasting	Prediction target not carbon-related
Scopus	107	Erlangga and Cho (2025)	Power demand prediction	Prediction target not carbon-related
Scopus	111	Alfayan and Hagras (2025)	Future kWh energy consumption prediction	Prediction target not carbon-related
Scopus	112	Novielli et al. (2025)	Soil respiration sensitivity (Q10) prediction	Prediction target not carbon-related
Scopus	113	Faisal et al. (2025)	Evaluation, clustering, and prediction of city-level carbon trading strategies	Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)
Scopus	117	Shmuel et al. (2025)	Lightning-ignited wildfire prediction and risk estimation	Prediction target not carbon-related
Scopus	118	Tripathi et al. (2025)	Customer retention forecasting	Prediction target not carbon-related
Scopus	119	Bhagat et al.	Material ranking, life cycle sustainability assessment, and optimization	Prediction target not carbon-related
Scopus	120	Al Mazroa (2025)	Multi-horizon load and price forecasting with carbon-aware scheduling	Prediction target not carbon-related
Scopus	121	Venkatachalam et al. (2025)	BTE, BSFC, EGT, NOx, CO, HC, and smoke prediction with blend/load optimization	Prediction target not carbon-related
Scopus	122	Zhao et al. (2025)	Global burned-area prediction followed by wildfire carbon-emission projection	Carbon emissions derived indirectly or not the XAI-explained target
Scopus	123	Altiary & Hussain (2025)	Systematic review of explainability in carbon price forecasting systems	Review or non-original article
Scopus	124	dos Santos et al. (2025)	CO_2 conversion and CH_4 selectivity prediction for Ni-based catalysts	Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)
Scopus	125	Belkadi et al. (2025)	Next-day methane-emission forecasting	No XAI applied

Database	No.	Author(s)	Prediction task	Exclusion reason
Scopus	126	Lu and Li (2025)	Annual and monthly building energy consumption and site EUI prediction; building-stock clustering	Prediction target not carbon-related
Scopus	127	Hussain and Soliman (2025)	Compressive strength, flexural strength, and fracture energy prediction	Prediction target not carbon-related
Scopus	128	Iwakin et al. (2025)	Optimal power-dispatch prediction for distributed energy resources	Prediction target not carbon-related
Scopus	129	Du et al. (2025)	NH_3 concentration and energy-efficiency prediction	Prediction target not carbon-related
Scopus	134	Yun et al. (2025)	Effluent TN concentration and energy-consumption prediction/optimization	Prediction target not carbon-related
Scopus	135	Zhang et al. (2025)	High-resolution daytime on-road CO_2 concentration and CO_2 increment (ΔCO_2) prediction	Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)
Scopus	136	Goto et al. (2025)	Electric mileage ratio (EMR) prediction and factor decomposition	Prediction target not carbon-related
Scopus	138	Ciobotaru et al. (2025)	Air-compressor component condition and fault classification	Prediction target not carbon-related
Scopus	139	Miao et al. (2025)	Compressive-strength prediction	Prediction target not carbon-related
Scopus	140	Ukwuoma et al. (2025)	Hydrogen (H_2) production prediction	Prediction target not carbon-related
Scopus	141	Escribano et al. (2025)	Operational-state classification: Activation, Reaction, Irregularity	Prediction target not carbon-related
Scopus	142	Paramasivam et al. (2025)	H_2, CO, LHV and CGE prediction	Prediction target not carbon-related
Scopus	143	Yousef et al. (2025)	Long-term wind-speed forecasting	Prediction target not carbon-related
Scopus	144	Kiouvrekis et al. (2025)	Shaft-power estimation/prediction	Prediction target not carbon-related
Scopus	145	AlShammari (2025)	Coefficient of Performance (COP) prediction and optimization	Prediction target not carbon-related
Scopus	146	Zhang et al. (2025)	High-temperature stability and low-temperature crack-resistance prediction	Prediction target not carbon-related
Scopus	147	Tieng et al. (2025)	Facility efficiency classification and energy-consumption prediction for energy saving and carbon reduction	Prediction target not carbon-related
Scopus	148	Zhao et al. (2025)	Hourly gross transaction value (GTV) / revenue forecasting	Prediction target not carbon-related
Scopus	150	Li et al. (2025)	HEV engine operating-state (ICE on/off) classification	Prediction target not carbon-related
Scopus	154	Tudor et al. (2025)	Share of fossil fuels (SFF) forecasting through 2030	Prediction target not carbon-related
Scopus	159	Olawumi & Oladapo (2025)	Energy-demand/consumption prediction and emission-reduced energy allocation	Prediction target not carbon-related
Scopus	160	Mungan & Arpa (2025)	Vehicle CO_2-emission prediction	No XAI applied
Scopus	161	Elbasi et al. (2025)	Energy-efficient crop-yield forecasting and resource management	Prediction target not carbon-related
Scopus	162	Inqiad et al. (2025)	Density, shrinkage and strain prediction of soilcrete	Prediction target not carbon-related
Scopus	163	Das et al. (2025)	PTC outlet-temperature prediction	Prediction target not carbon-related
Scopus	164	Fuqaha et al. (2025)	Concrete compressive-strength prediction	Prediction target not carbon-related
Scopus	165	Bhambare et al. (2025)	Review of AI applications for environmental monitoring, energy optimization and emission reduction	Review or non-original article
Scopus	166	Ersöz et al. (2025)	Li-ion battery performance prediction during drone flights	Prediction target not carbon-related
Scopus	168	Asim et al. (2024)	Review of price, demand, load, carbon-emission and supply forecasting	Review or non-original article
Scopus	169	Bao et al. (2024)	CO_2 conversion ratio, CH_4 selectivity and exergy-efficiency prediction	Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)
Scopus	170	Mahamoodally et al. (2024)	Imbalanced occupancy classification with domain adaptation	Prediction target not carbon-related
Scopus	172	Zhang et al. (2024)	Review of EV energy-consumption prediction and modelling	Review or non-original article
Scopus	173	Liu et al. (2024)	Air-quality status classification and CEP rule extraction	Prediction target not carbon-related
Scopus	174	Zhang et al. (2024)	On-road CO_2 concentration prediction and spatial mapping	Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)
Scopus	175	Guo et al. (2024)	Compressive-strength prediction + multi-objective low-carbon material design	Carbon emissions derived indirectly or not the XAI-explained target

Database	No.	Author(s)	Prediction task	Exclusion reason
Scopus	176	Shokrollahi et al. (2024)	CO_2 solubility prediction in brine solutions	Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)
Scopus	177	Ngeni et al. (2024)	Bike-sharing station demand/capacity prediction	Prediction target not carbon-related
Scopus	178	Elmousalami et al. (2024)	Project cost, cost variance, duration and duration-variance prediction	Prediction target not carbon-related
Scopus	179	Arbaoui et al. (2024)	Battery State of Health (SOH) forecasting and abnormal degradation detection	Prediction target not carbon-related
Scopus	181	Alyami et al. (2024)	Compressive-strength prediction of rice-husk-ash concrete	Prediction target not carbon-related
Scopus	182	Nallakaruppan et al. (2024)	Solar AC power-output prediction	Prediction target not carbon-related
Scopus	184	Wang et al. (2024)	Aeration/air-demand prediction for low-carbon WWTP operation	Prediction target not carbon-related
Scopus	186	Lee et al. (2024)	Fuel-consumption prediction and derived CO_2-emission estimation	Carbon emissions derived indirectly or not the XAI-explained target
Scopus	188	Lee (2024)	Fill-factor prediction of non-fullerene organic solar cells	Prediction target not carbon-related
Scopus	189	Sison et al. (2024)	H_2 yield and char-yield prediction	Prediction target not carbon-related
Scopus	190	Vellayappan et al. (2023)	Syngas ratio, CH_4 conversion and CO_2 conversion prediction	Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)
Scopus	194	Fatahi et al. (2023)	Kiln feed-rate and ID-fan-current prediction	Prediction target not carbon-related
Scopus	195	Shah et al. (2022)	28-day compressive-strength prediction	Prediction target not carbon-related
Scopus	196	Heo et al. (2022)	Offshore wind-power forecasting and downstream net-zero carbon assessment	Carbon emissions derived indirectly or not the XAI-explained target
Scopus	197	Moraliyage et al. (2022)	Building energy-consumption prediction and energy-savings estimation	Prediction target not carbon-related
Scopus	198	Wenninger et al. (2022)	Annual building Final Energy Performance (FEP) prediction	Prediction target not carbon-related
Scopus	199	González-González et al. (2022)	Bank-transaction activity classification followed by carbon-footprint estimation	Carbon emissions derived indirectly or not the XAI-explained target
Scopus	200	Mouakher et al. (2022)	Daily, weekly and monthly energy-consumption forecasting	Prediction target not carbon-related
Scopus	201	Takalo-Mattila et al. (2022)	Steel surface-defect classification	Prediction target not carbon-related
Scopus	202	Labe and Barnes (2021)	Year/decade prediction from near-surface temperature maps under different climate forcings	Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)
WoS only	13	Hou et al. (2026)	Seasonal block-scale land-surface-temperature prediction	Prediction target not carbon-related
WoS only	16	Do et al. (2026)	Review of ML applications in solar forecasting, optimization, fault detection and energy management	Review or non-original article
WoS only	20	Tahera et al. (2026)	Mechanical and durability performance prediction with LCA-based embodied-carbon assessment	Carbon emissions derived indirectly or not the XAI-explained target
WoS only	33	Şişman & Yavuz (2026)	EV adoption-intention classification	Prediction target not carbon-related
WoS only	53	Akdoğan (2025)	Prediction of voluntary carbon-disclosure behavior	Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)
WoS only	84	Chen & Jia (2024)	Chinese coal-consumption and coal-price-index prediction	Prediction target not carbon-related

Reports Not Retrieved (n = 10)

Full texts were not accessible through institutional access or open sources; these records were not assessed for eligibility.

Database	No.	Author(s)
Scopus	40	Ai and Lu (2026)
Scopus	47	Zhang et al. (2026)
Scopus	48	Xu et al. (2026)
Scopus	64	Chen et al. (2026)
Scopus	80	Luo et al. (2026)
Scopus	86	Ma et al. (2026)
Scopus	97	Zheng-Yi et al. (2026)
Scopus	102	Guo et al. (2026)
Scopus	149	Wenjie et al. (2025)
WoS only	12	Kim et al. (2026)

Summary of Exclusion Reasons

Counts correspond to the PRISMA 2020 flow diagram (Fig. 1).

Exclusion reason	Scopus	WoS-only	Total
Prediction target not carbon-related	92	3	95
Carbon-related target outside scope (e.g., CO_2 capture, conversion, concentration, temperature)	12	1	13
Carbon emissions derived indirectly or not the XAI-explained target	10	1	11
Review or non-original article	8	1	9
No XAI applied	3	0	3
Total	**125**	**6**	**131**